MILITARY SPECIFICATION
CARTRIDGE, CALIBER .30, BALL, FRANGIBLE, M22

1. SCOPE

1.1 This specification covers Cartridge, Caliber .30, Ball, Frangible, M22 for training purposes.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

Munitions Command

C7640951	- Cartridge, Caliber .30, Ball, Frangible, M22
C7643674	- Classification of Cartridge Case Defects
D7553074	- Packing and Marking; Cartridges, Caliber .30, Ball, Frangible, M22, Cartons, Can, Box, Ammunition, M22
IEL-7640951	- Inspection Equipment List for Cartridge, Caliber .30, Ball, Frangible, M22

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2.1 (Cont'd)

PUBLICATIONS

AMCR 700-700, Vol. III - Manual of Test Methods for Small Arms Ammunition

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with the requirements specified on Drawing C7640951, referenced specifications and the following:

3.2 Residual stress.- The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.3 Velocity.- The average velocity of the sample cartridges, conditioned at $70^{\circ} \pm 2^{\circ}$ Fahrenheit (F), shall be 1320 feet per second (ft/sec) plus or minus 30 ft/sec, at 53 feet from the muzzle. No individual velocity shall exceed 1500 ft/sec.

3.4 Waterproof.- The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 5 pounds per square inch for 15 seconds.

3.5 Accuracy.- The average of the mean radii of all targets of the sample cartridges fired at 100 yards shall not exceed 2.0 inches.

3.6 Perforation.- The bullet of the sample cartridge shall not perforate a 3/16 inch thick aluminum plate at 25 yards. The plate shall be Dural 2024 T4 (or equivalent) with a Brinell hardness of 105 to 125 under a 500 kilogram load.

3.7 Function and casualty.- The cartridge shall function without casualty.

3.8 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects (Case).- The cartridge case shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

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3.8.2 Material defects (Bullet).- The bullet shall not be chipped and shall be free of dents, scratches, voids and seams, broken points and other defects.

3.8.3 Foreign matter.- The cartridge shall be free of corrosion, discolorations, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 500 cartridges. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

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4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model as well as with a lot number and the manufacturer's identification as assigned by the procuring agency. Each lot shall be further identified by a Federal Stock Number assigned by the procuring agency.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non-conforming cartridges shall be rejected.

4.3.2.1 Classification of defects shall be as specified in Table I.

TABLE I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded or stained, if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case		X		
7	Perforated case	X			
8	Draw scratch				X
9	Scratch			X	
10	Beveled underside of head		X		
11	Case mouth not crimped		X		
12	Scaly metal				X
13	No chamfer on head (rim)		X		
14	Fold			X	
15	Wrinkle			X	
16	Buckle			X	
17	Bulge			X	
18	Illegible or missing head stamp			X	

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4.3.2.1 (Cont'd)

TABLE I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Case (Cont'd)				
19	Defective head			X	
20	Defective mouth			X	
21	No visible evidence of mouth anneal		X		
	Bullet				
22	Dent			X	
23	Scratch			X	
25	Loose bullet		X		
27	Voids and seams		X		
28	Chipped			X	
29	Broken point		X		
30	Blunt point			X	
	Primer				
32	No primer		X		
33	Cocked primer		X		
34	Inverted primer		X		
35	Loose primer		X		
36	Nicked or dented primer			X	
37	No waterproofing material (primer pocket joint)			X	
38	Defective crimp			X	
	Gaging				
39	Total length		X		
40	Cartridges profile failure (requiring more than 20 lbs. dead weight to insert in profile and alignment gage)		X		
41	Diameter of extractor groove, max.		X		
42	Diameter of extractor groove, min.			X	
43	Diameter of head		X		
44	Thickness of head		X		
45	Length to shoulder datum, min.		X		
46	Depth of primer		X		
47	Weight 2/	X			

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4.3.2.1 (Cont'd)

1/ Refer to Cal. .30 section of MIL-STD-636 for visual standards for defect Nos. 1 through 38 except defects 27, 28 and 29. In the event of conflict between Table I of this specification and MIL-STD-636 as to defect classification, the classification specified in Table I shall apply.

2/ Each light or heavy weight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain from 6 to 18 grains of propellant inclusive shall be counted as a major defect. Any cartridge containing less than 6 or more than 18 grains of propellant shall be classed as a critical defect.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges selected shall be thoroughly mixed before being divided into test samples.

TABLE II

Test	Number of Cartridges Ambient Temperature	Requirement Paragraph
Mercurous Nitrate 1/	20	3.3
Velocity 2/	20	3.4
Waterproof 4/	20	3.5
Accuracy 2/	90	3.6
Perforation 1/	50	3.7
Function and Casualty 3/ Gun, Machine, Cal. .30 M37	200	3.8

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the initial test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

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2/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample, consisting of double the quantity of cartridges in the first test. Failure of the cartridge in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

3/ The initial production sample or production lot shall be rejected when function and casualty defects plus all other firing defects exceed the acceptance number for the cumulative sample of Table III. If the number of defects found in the first test exceeds the acceptance number of the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantity specified under function and casualty test, shall be fired in all the service weapons specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred in the first test. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing the second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance numbers for the cumulative sample, the lot shall be rejected.

4/ Failure of nine or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than four but less than nine cartridges fail in the initial test, a second sample consisting of double the number of cartridges in the first sample may be tested. The lot shall be rejected if, in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

TABLE III

Defects	Acceptance Number: First Sample	Acceptance Number: Cumulative (First & Second Samples)
1. Misfire	1	2
2. Bullet remaining in bore 1/	0	-
3. Primer leak		
a. Perforation in firing pin indent in primer cup Machine guns	4	9
b. Escape of gas through primer cup other than 3a	1	2

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4.3.3.2 (Cont'd)

TABLE III (Cont'd)

Defects	Acceptance Number First Sample	Acceptance Number Cumulative (First & Second Samples)
3. Primer Leak (cont'd)		
c. Escape of gas around primer cup more than 50% of periphery	3	6
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
e. Primer remains in pocket but is physically loose	2	4
4. Case Casualties		
a. Longitudinal split 2/		
(1) Neck and shoulder (I & S)	7	13
(2) Body (J)	2	4
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 2/		
(1) Partial, Shoulder or body (S, J or K)	2	4
(2) Partial Head (I)	0	1
(3) Complete	0	1
5. Failure to Extract	0	1
6. Bullet disintegrating before striking target or terminal	1	2

1/ No second sample permitted. Lot shall be rejected.

2/ For location of defects indicated by numbers in parenthesis, see Drawing C7643674.

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a dummy, blank or grenade cartridge shall be classed as a major defect. Occurrence of any type other than those listed shall be classed as a critical defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

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4.3.5 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-7640951 except as provided in 4.3.

4.4 Test methods and procedures.

4.4.1 Mercurous nitrate.- Test shall be conducted in accordance with AMCR 715-505, Vol. 3.

4.4.2 Velocity.- Test shall be conducted in accordance with AMCR 715-505, Vol. 3.

4.4.3 Waterproofness.- The test shall be conducted in accordance with AMCR 715-505, Vol. 3.

4.4.4 Accuracy.- The test shall be conducted in accordance with AMCR 715-505, Vol. 3.

4.4.5 Perforation.- This test shall be conducted in accordance with AMCR 715-505, Vol. 3, Chapter 1, Section 12, with the following exceptions:

a. Velocity is not measured for the test cartridges.

b. To qualify as a record shot, bullets must strike plate at least 2" apart or 2" from the edge of the plate.

c. Number of cracks and their extent are reported.

4.4.6 Function and casualty.- This test is to be fired, single shot, 100 rounds in each of two M37 machine guns. The weapons and ammunition shall be at room temperature at the beginning of the test. The test shall be conducted in accordance with AMCR 715-505, Vol. 3.

4.4.7 Defect penalty.- In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing - Level A (Worldwide Shipment).- The cartridges shall be packed in accordance with Drawing D7553074.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with drawings cited in 5.1.

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6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type of packing.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1305-A544

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